# NoMoTown

## An agent-based model of transport mode choice

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*In most cities, cars remain the dominant mode of transport. This is a huge problem not only because of obvious effects such as congestion and pollution, but also because it causes health issues for commuters themselves which lead to further costs for the community. In our work, we have developed an agent-based simulation which offers mitigation strategies and tries to propose realistic lines of action for transport modes to more sustainable modes. Our approach can import from GIS or (raster) maps, thus acting as a planning tool for urban planners and city administrators; we also included the possibility for generating theoretical / idealized cities, as a testbed and theoretical tool for instructing policy makers. Our goal is to find an equilibrium between individual freedom in transport choice, financial effort required for maintaining the overall transport system and the health of the whole population.*

**Keywords:** *Agent-Based Simulation; Urban Dynamics; Multimodal Transport; Sustainability.*

## INTRODUCTION

The term "MoTown" is a portmanteau of *motor* and *town*, typically used for large cities that predominantly rely on individual motorized transport for which they provide the infrastructure; other means of transport - such as public transport and bicycle traffic - are not supported to the same extent as automotive transport. This is bad for the environment, but as well for the health of the whole population as we argue: Lack of exercise and exposure to pollutants and fine dust leads to a high amount of cardiovascular diseases (most notably heart attacks and strokes), type-2 diabetes and cancer (U.S. Centers for Disease Control and Prevention, n.d.). In our work, we have developed an agent-based simulation which offers mitigation strategies and tries to propose realistic lines of action for shifting transport modes to more sustainable and healthy modes. Our approach can import from GIS or (raster) maps, thus acting as a planning tool for urban planners and city administrators; we also included the possibility for generating theoretical / idealized cities, as a testbed and theoretical tool for instructing policy makers. Our goal is to find an equilibrium between individual freedom in transport choice, financial effort required for maintaining the overall transport system and the health of the whole population.

In more detail, we have structured our work in the following fashion:

- We first take a look at our model (section "NoMoTown agent-based model") and give an overview of its parametrization.
- In the next step, we apply the model in order to find specific lines of action that enable us to shift transport modes to more healthy and sustainable ones (section "Lines of Action"). In

that section we also discuss the sensitivity of the system towards parameter changes and the "point of no return" after which we cannot reach a sensible equilibrium between sustainable transport, maintenance costs and population health.

As mentioned, our approach can act as a basis for both theoretical discussion as well as for planning; we discuss some aspects of this (section "Discussion") before concluding.

## RELATED WORK

Whole-society models have been made popular by the 1972 Club of Rome study "The Limits of Growth" (Meadows et al. 1972) which had a high influence on popular culture in form of games such as "SimCity" (Wright 1989), but also in scientific fields such as the social sciences (Epstein and Axtell 1996).

In urban planning, modeling societies with regards to traffic mode changes has been done to a much lesser extent: Bori et al. (2006) have modeled transport choice by means of behavioral models for the sake of optimizing transport systems. Aschwanden et al. (2010) have looked at generated emissions in procedurally generated cities. Hartl and Wurzer (2013) have considered options for switching transport modes to sustainable ones for the specific case of Vienna/Austria. A general survey of transportation in agent-based modelling has been given by Wise et al. (2016). None of the aforementioned approaches can serve both for theoretical considerations as well as practical planning tool, which is a clear bonus of our approach.

Some other models are more sophisticated and/or have a higher degree of realism or complement the work we do: For example, Basu et al. (2018) have looked at automated mobility-on-demand vs. mass transit; Kim and Batty (2011) have looked at microeconomics and spatial policy simulation; Zhao and Peng (2012) have investigated land use change using a cellular/agent-based model.

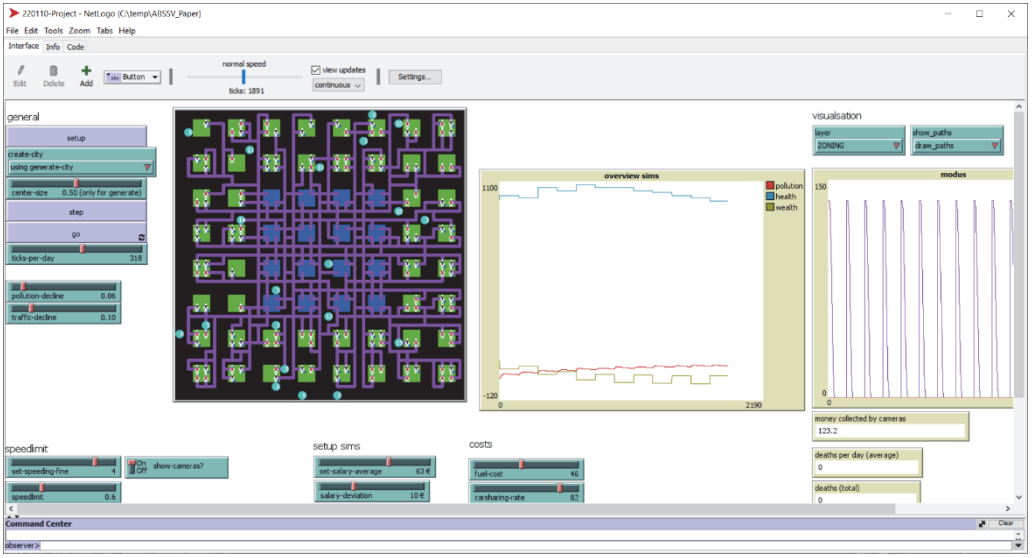


Figure 1
View of the NoMoTown model (implemented in NetLogo [Wilensky 1999])

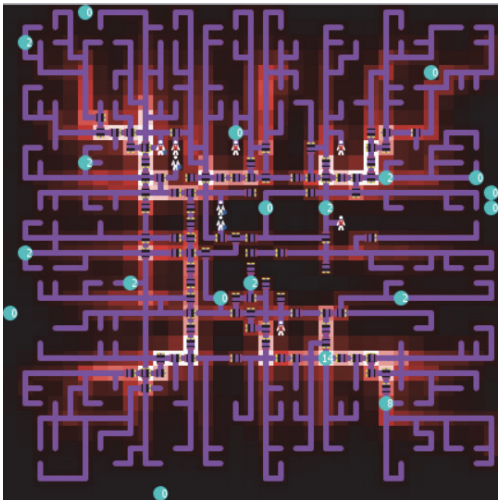

Figure 2
Traffic and commercial/industrial activity generate pollution which is subject to dissipation after a certain amount of time.

## NOMOTOWN AGENT-BASED MODEL

NoMoTown is an agent-based model which executes in a cellular automaton (see figure 1). The cells are labeled as having residential, commercial, industrial, or no usage. We furthermore attribute different population densities and incomes for each cell. Available means of transport (cars, e-cars, bikes) and existing traffic infrastructure (roads attributed with allowed traffic modes) are modeled as agents which sit on top of the cell raster.

A real city configuration can be read from a GIS package by the way of ESRI Shapefiles, or by importing color-coded raster images (PNGs). Furthermore, we also have the opportunity to generate artificial cities which can serve as a theoretical test-bed. Our current implementation allows for 1.) typical "MoTowns" with a center of parametrizable size, containing commercial and industrial areas, suburbs containing housing, and 2.) uniform distribution of residential, industrial, and commercial areas.

The simulation core is occupied with moving inhabitants from the housing areas to the commercial/industrial areas in the morning, and back again in the evening. Depending on the mode of transport and commercial/industrial activity, pollution is generated and dispersed after a certain amount of time (see figure 2).

Pollution decreases the health of simulated agents, which is said to be critical after a certain preset threshold. Vice versa, we increase health when there is lack of pollution.

Each simulated agent has a certain amount of money which is increased by doing work and decreased with regards to the chosen transport mode (read: *fuel* in the case of own car use; carsharing tariff; no decrease for bike). As further economical aspect, we have implemented speeding fines, which are simulated as a stochastic process which all cars can be subjected to.

The outputs of the model are recorded in the form of two graphs (refer to figure 3):

- We plot the health and wealth of the population versus the pollution generated by traffic and commercial/industrial activity (left in figure 3).
- We plot the mode of transport being used (right in figure 3), which can be either using a regular car, using carsharing with e-cars, or using a bike.

The core of our search for equilibrium lies in the mode switching rule outlined in algorithm 1: If an agent's health is below a certain limit, then a switch to a more sustainable mode occurs. Depending on the agent's wealth, either a switch to carsharing (using e-cars) or to bikes is automatically done.

The effects of this rule are immediately visible in the left part of figure 3 (indicated by an arrow): As the overall health shown in the left part of figure 3 decreases beyond a threshold, other modes of transport are sought for (right part of figure 3).

## LINES OF ACTION

We have used the model to investigate a number possible scenarios that are realistic for mode switching, which we want to bring here and discuss later on (section "Discussion").

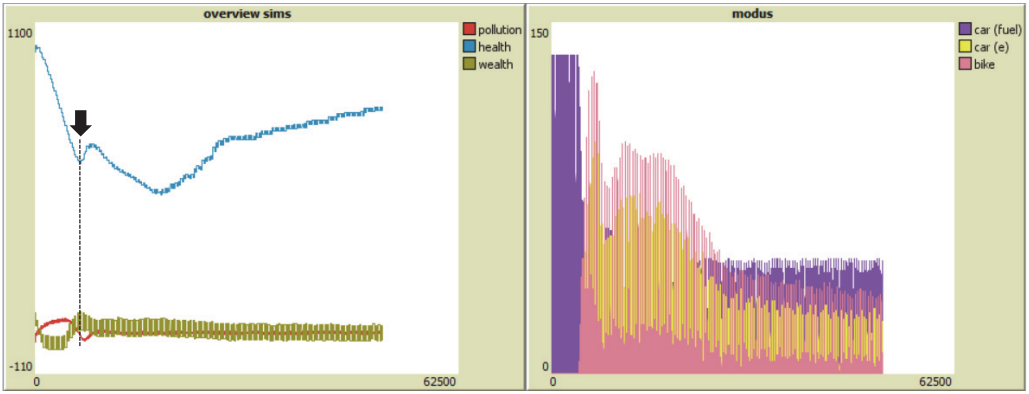


Figure 3
Output of the NoMoTown model shown as graphs (left: pollution vs. health vs. wealth; right: used modes of transport). Notice the automatic mode switch that is indicated by the arrow.

**Algorithm 1: ModeSwitch(*agent*)**

```
 if agent.health > critical-health and
   agent.money >= fuel-costs
 then
     use car and and be subjected to fuel costs
 else
     if agent.health <= critical-health and
       agent.money >= carsharing-cost
     then
       use e-car in carsharing and pay carsharing-cost
     else
       use bike
     end
 end
```

## Mixing workplaces with residential areas

A uniform arrangement of residential, commercial and industrial areas would trivially take the initial planning problem ad absurdum, if the population is also free to choose the nearest workplace and does thus not need to commute. One could argue that this must be the overall goal of sustainable development which is hard to achieve but would nevertheless be the optimum… but interestingly such a mix would not necessarily lead to dramatic health improvements among the population, because the closeness of activities also leads to less exercise. It does, however, entail less overall pollution, so there is a certain subtle balance between to be gained here.

A more realistic route to success which is already being taken is to focus on the belt between the city core and the suburbs (see figure 4) and to further impose restrictions on the city core, as given in the next subsection.

## Decrease throughput in dense areas

We implemented speed limits in the city center (see dark grey roads as opposed to light grey roads in figure 4 which act as weights during path finding (home→work, work→home). This speed limit is tested at a parametrizable number of randomly distributed control points (see dots in figures 1 and 2); speeding is also governed by a stochastic and parametrizable process (plus/minus speed). At the moment, speeding fines are collected but not used to subsidize more sustainable modes of traffic, which is a route for further work (see "Discussion").

If taken to the extreme this measure will distribute traffic in a high-volume belt around the

city center, with incursions only happening when absolutely needed and there is no other way (i.e. close to the workplace).

Another possible decrease of attractiveness for dense areas would be the inclusion of a parking/tolling model which is well beyond scope of this paper, but a really good point for extension in future work since it reflects what is being done today quite nicely (also see next section for other mitigating factors).

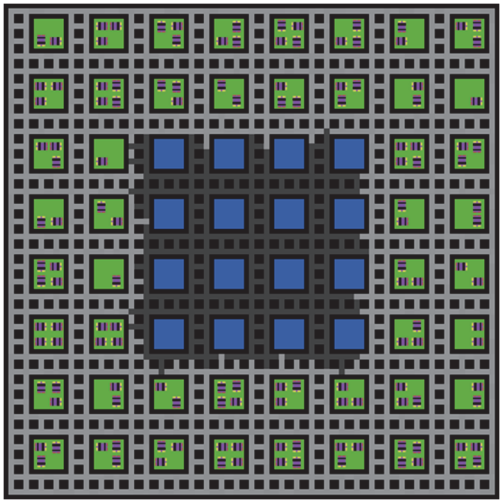

Figure 4
Decreasing throughput in dense areas (dark gray roads in the city core vs. light grey roads around that).

### Increase prices for unsustainable modes

Obviously, transport price (i.e. fuel or fare in the case of carsharing) governs the individual choice more than "soft" measures such as speed limits. However, and contrary to intuition, our simulation shows that people living in the suburbs are able to endure long commutes and still be healthier than people living closer to the city core, since pollutants are emitted along the main commuting routes and do not appear in the suburbs, or dissipate quickly because the traffic density is minor.

Would it make sense to tax people for clean air, knowing that renting prices in the outer limits are much less then in the city core? Often, this question cannot be answered since city core and suburbs belong to a different political entity (read: NUTS-2 region [NUTS-2 Wikipedia]) and it is exactly this inequality that the suburbs see as advantage, while the city itself sees it as injustice. In that light, this paper does not even try to approach this extension point in future work (without discouraging others from pursuing that path).

## DISCUSSION

Our model does not take all aspects of reality into account. For example, mode choice does not only depend on economic viability (even though this is important), but is also significantly influenced by distance. Weighting by distance has been initially included but we found out that transport choices are modeled more boldly when this is not the case, at the price where we cannot model mega-cities where this is strictly unviable. To be clear, our target cities have at most 2 million inhabitants, so this does not apply to mega-cities in southeast Asia such as the Shenzhen/Hong Kong region.

Furthermore, we could have added a fourth class named "public transport", which would have been almost the same as "carsharing via e-cars" (there is a tariff, but public transport causes less pollution than individual motorized traffic). The reason why we chose to remain with individual transport is the reference to MoTown, i.e. automotive cities which have next to no public transport infrastructure. This is true for a lot of cities on the European continent, as well as (obviously) a lot of cities in the Americas and the Middle East.

The next concern is about applicability for real cities: Even though it is possible to import GIS data, this would have to be prepared as areas with residential, commercial or industrial attribution (and critics may rightly argue that this differentiation was introduced by SimCity, not real city planning); but what if we let these justified concerns out of the argument for the moment and assumed that we already had the ideal case of a mixed zoned city in which residential, commercial and industrial areas

were equally spaced? Clearly, this approach still assumes that this would not be the case, as cities are aggregated (in a cellular automaton) for their "predominant type" of cell and there is no mixed industrial/residential/residential type; the model simply assumes these are separated. An obvious possibility to achieve this separation would be to increase spatial resolution, but at the same time the model demands that we are not stepping down to the block level but remain larger, in order to catch large-scale effects that are hard to capture in a bottom-up fashion. Counter-argument: Why apply this model at all if the city is already optimal?

Another point for discussion is the application of only individual transport switch rules, but not global ones: For example, the initial model disallowed individual motorized traffic if the mean health of x percent the population was below a certain threshold y. But what does that mean? Are we in a totalitarian dictatorship in which the individual choice is superseded by the need of the many? We have tested this route of though in our simulation model, and results were quite expectable (namely that sustainable modes were easily achieved by oppression), but we have obviously not continued. We believe that oppression reflects inadequacy or lack of empathy, and that before such measures individual choice must be considered. Individual choice not as a sign of egoism, but out of necessity (e.g. home→work and work →home; home in the suburbs, out of financial considerations).

From a simulation standpoint, this model is very sensitive to changes in all of its 19 parameters (implicit assumptions raising that to around 25). We are not neglecting that, however at the moment we are setting "sensible" values and iterating through them, however, a massive parameter study would surely expose some additional points for enquiry (25!=1.551121e+25 possibilities and that would be only if these would be if these were binary, so there is surely a long way to go). To our defense, we must say that we have developed the model in a manner that suits the argument, and likely the main routes have been taken (and still some inconsistencies need to be discovered, whether they'd be from the context or the implementation). At the moment we are sincere that we captured the main dynamics and this paper is at least complete in that meaning, but we cannot guarantee that there additional points in our research that we did not devote our attention to.

Every system has its breaking point. Adjustments to the impact of pollution, dissipation of the latter or simply the salary each individual gets by working can alter the results of this model significantly, but this is not the argument that we want to make. The real argument is the "point of no return" after which the health of the population (with dependencies on income and spatial location ~= chosen workplace) simply does not work out, and the system goes into chaos. The counter argument in that case is that the planning effort is targeted at achieving an orderly behavior of the city vs. the inhabitant, and vice versa and orderly behavior of the inhabitant vs. the city. We certainly measure the effects of our model as per U.N. Sustainable Development Goal 3 [SDG-3 (Health)], and are thus biased. Others might employ different metrics (e.g. economic ones, fully warranted) and reach other conclusions. This is what the discussion is about.

Concerning the latter point, we must also confess that the redistribution of wealth [from speeding fines, speaking in concrete terms just for this model] was not being considered, but since the choice of subsidy is also complicated we have not chosen to go further with this; which, of course, is not an oversight but a decision, and certainly this will be addressed in further work. This does entail into sponsoring public transport in MoTown as well, which is why (for this paper) we have excluded the point from active inquiry.

Another point for extension would be an application of this method using an actual city, using GIS data. Furthermore, as one reviewer of this paper correctly noted, it would also be nice to include mixed-use areas which are not fully implemented at this time (commercial/industrial exists, but not residential/industrial or residential/commercial).

## CONCLUSIONS

We have presented a model called NoMoTown which shifts (commuter) transport to more sustainable modes. The main result is that we should monitor health of individual citizens for determining transport choice, which is something that is generally done (e.g. eye test for older persons) but hardly utilized in practice. We argue that these efforts need to be put on the main agenda, since citizens should be encouraged to focus on more sustainable modes of transport and city planners should put more opportunities for nearby work into the hands of people.

## ACKNOWLEDGEMENTS AND AUTHORSHIP INFORMATION

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